

Using Machine Learning Techniques to Detect Malware in IoT Network Traffic

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Abstract—IoT devices are computing devices with the ability to connect over a network. Such devices offer many services and are exponentially increasing in production and use. However, the increase in IoT devices in the last decade has introduced a new vector for hackers to develop and target such systems. Various vulnerabilities and weaknesses in such systems even made infecting IoT devices with malware, gaining unauthorized access and including the device in a larger DDOS attack relatively simple. In our research, we use various machine learning models on a labelled IoT dataset that contain both malicious and benign IoT network traffic. Furthermore, we measure our classification results by various performance metrics such as training and testing accuracy and F1 scores. According to the conducted analysis, Decision tree classification resulted in an average accuracy of about 0.972.

Keywords—IoT, Malware, Machine learning, IoT Security

I. INTRODUCTION

Internet of Things (IoT) is defined as “the network of physical objects— “things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet” [5]. IoT devices are a promising new and lightweight technology that can revolutionize the current ecosystem of many industries. The applications of such devices can be seen in smart cities, smart bikes, fire alarms, watches, and door locks for the purpose of enhancing the quality of different services. Despite the compelling services and designs offered in IoT devices currently sold in the market, IoT is still a new field in computer technology under development. As a result, IoT networks and applications are vulnerable against cyber-attacks and are being increasingly subjected to various types of security threats. Attackers are constantly able to disrupt and control IoT devices using malware and other attack vectors such as DDOS (distributed denial of service). Since IoT devices have only enough computing power to do what they’re created to do, they cannot support programs such as antiviruses to get rid of malware, worms, and botnets. Mirai botnets, according to Cloudflare, is a malware that infects smart or IoT Devices that run on ARC processors [1]. These malwares, after infecting the devices, target specific organizations depending on the hacker and start DOS attacks, send spam emails, perform click fraud, solve CAPTCHAS to imitate humans, and steal credit card information. It was estimated that the Mirai attack infected 600,000 IoT devices and disrupted services such as Amazon and Twitter [3].

Moreover, ever since the creation of the Mirai botnet, the malware is mutating because the original creators of the botnet

published the source code on the internet to not get caught. This project will mainly focus on Mirai botnet detection since most other botnets are derived from it. Therefore, the network traffic generated from the IoT devices can be used to better understand and study the patterns for such attacks and prevent them before causing damage to the privacy of users and an outbreak to many companies and services. The goal of such project is to classify various and different malware and attacks that infect various IoT devices. The classification of different malicious activities will aid in detecting patterns to how an attacker performs different attack stages. It is important to understand that we all use IoT devices such as smart speakers, smart home devices, smartwatches, smart door locks and medical sensors. However, without the strong and strict security policies that IoT devices lack, they are left to many malicious attackers to invade and disrupt, steal, and compromise people’s lives and privacy

II. NEED

Using machine learning is necessary when human knowledge and expertise cannot be used in such complicated tasks. Security breaches and techniques used in IoT devices are constantly changing and dynamic, which is why the use of machine learning based detection system is the need of the hour. Moreover, IoT devices produce a good amount of data which includes IP address, MAC address, and packets and through various ML techniques a smart intrusion system can be created. The data generated by such IoT devices can be used to take better informed decisions by the system to prevent the attack. The aim of this project is to detect, analyze and study the patterns used by attackers when performing attacks on IoT devices. There are certain features that are to be selected from this dataset, for example from a certain classes of IP address, ports, and protocols, that will allow us to maximize the accuracy of the classification of the attack. Through research, we will create an intrusion detection system for IoT devices using the common machine learning techniques such as K-nearest approach, Random Forest, and Naïve Base. The different set of capture files and features included in the dataset will be combined and reduced to generate the best classification results.

III. DATASET

To achieve the proposed plan, the IoT-23 dataset, which consists of IoT network traffic collected in the Stratosphere Laboratory [4], will be used. This dataset contains 23 different

captures or scenarios of different IoT network traffic. In each scenario a different type of malware/attack was tested on IoT devices. Each capture represents an execution of a specific malware in the IoT device, and the packets collected over the network when the device was connected and infected. Each capture contains the original pcap file and the conn.log.labeled which was obtained by executing the Zeek network analyzer on the pcap files. The relevant file that will be used is the conn.log file. A part of the dataset will be used to train the model. 4 captures scenarios will be processed and worked with. The Amazon Echo Capture (CTU-Honeypot-Capture-5-1) contains 1383 entries and the Mirai capture (CTU-IoT-Malware-Capture-34-1) contains 23145 entries. Also, (CTU-IoT-Malware-Capture-35-1) capture and (CTU-IoT-Malware-Capture-42-1) trojan captures were both used. The final dataset generated for this project will have a combined total of 23610 entries, with 5600 benign packets, 4870 C&C packets, 6760 DDOS packets and 6422 PartOfAHorizontalPortScan packets. Combining 4 captures will even the distribution of benign and malicious and yield better results in the classification phase

IV. LITERATURE REVIEW

The following section discusses some of the research conducted on the IoT-23 dataset using Machine learning models. Researchers developed Adept which is a distributed framework to detect and identify the individual attack stages in a coordinated attack. A data mining technique, frequent itemset mining was used to exploit the spatial and temporal characteristics of alerts received and extract patterns corresponding to attack stages. Then, malicious activity stages were classified using machine learning techniques. They use (k-NN), random forest (RF), and support vector machines (SVM). The classification performance was tested using fourfold cross validation in all the experiments performed, where they partition the data set into 75%: 25% training and testing. An accuracy of around 99% was achieved. [3]. In [7], researchers used the dataset to have a total of 32,382 benign data points and 1,676 malicious data points randomly extracted. Several ML algorithms were used such as: decision tree, random forest, multi-class decision forest, two-class neural network, and multi-class neural network. Information gain on the small dataset was calculated using the Waikato Environment for Knowledge Analysis. Their results were as follows: They compared the models generated for the small dataset and found that the best two are: models based on the performance metrics chosen are the multi-class decision forest and multi-class neural network models. The most consistently important features in terms of information gain found were: packet spacing, id_resp_p, resp_bytes, history – S, conn_state – SF, and resp_pkts. In [8], Researchers have implemented a mechanism for detecting network anomalies using the IoT-23 dataset. The algorithm contains: dataset selection, feature engineering, then a classifier in which the LSTM cells are feeding the final cell. For Dimensionality reduction, they use an auto-encoder of three hidden-layers that use the sigmoid activation function. The outcome is then passed to the

classifier, which is built on mLSTM classifier, DNN, and LSTM. They compared the accuracy of each of the classifiers based on 5 Folds. The average accuracy was 99.688% 99.844% 99.98% for DNN, mLSTM and LSTM respectively. Their results showed that compared to RF, SVM, and MLP used in machine learning, neural networks showed a better accuracy of classification. [8]. In Technique for IoT Cyber Attacks Detection Based on DNS Traffic Analysis (Bobrovnikova, Lysenko, et. Al , 2020), the research was able to detect that malicious IoT devices created more traffic than the benign IoT devices. To be able to detect malicious traffic from normal traffic, the researchers have figured out that they were successful in finding the bad traffic from the DNS Traffic analysis. The method used semi-supervised classification algorithms and were able to detect unknown IoT cyberattacks with a high efficiency of 96.06-98.01 percent with false positives ranging from 0.015-0.31%. [9]. In Machine Learning for Anomaly Detection in IoT networks: Malware analysis on the IoT-23 Data set (Stoian, 2020), the research investigates the usability in ML algorithms by detecting anomalies found within the data of the networks. It also examines ML algorithms that are successfully utilized in similar situations and compares it with its own data. The research implemented RF (Random Forest), Naive Bayes, and Multi-Layer Perceptron. The result of this research shows that the best option to detect the anomalies was the Random Forest which had an accuracy of 99.5%. [10]. The paper in [11] proposes a Delay Compensated Adam approach which is an asynchronous federated learning-based detection approach. After the training model is done, it is appended to a combination of theoretical analysis of convergence and practical results. Comparing the results of the research to a barrier-free asynchronous federated learning, the research method outperforms the barrier-free by 12.8% accuracy and 14% precision. [11]. In another research, the dataset consisted of 1,027,714 flows subsampled from approximately 65 million data points from the IoT-23 dataset. The methodology approach consisted of four classifiers including Random Forest, Naïve Bayes, Linear SVM, and Stochastic Gradient Descent. The most important features for the classification were: time of day, history, protocol, and count of origin bytes sent. Of these, time of day and volume of traffic coming from the same IP addresses are consistent for port scanning, infection, and distributed denial of service attacks. The results showed that the best performing algorithm was Random Forest with precision of 99.92%, accuracy of 97.45% and F1 score of 97.39%, whereas Linear SVM with a recall score of 99.90%. [12]. Another research used the ensemble approach which constitutes a team of models or base learners that work together in producing a better result. Furthermore, such methodology was divided into boosting, bagging and stacking techniques. In terms of boosting, the models used were C5.0 and GBM. C5.0 uses Information Gain value for attribute selection measure for building a decision tree and Binomial Confidence Limit as a pruning technique to obtain higher accuracy value. Whereas GBM, constructs new base learners which are maximally correlated with the negative gradient of the loss function, associated with the whole

ensemble. Next, in terms of boosting, the models used were Random Forest and Bagged CART. Random Forest consists of many decision trees where each one gives a class prediction and the class that has the most number of votes becomes the final prediction. Whereas CART, uses the Gini index as an attribute selection measure for building decision trees and cost complexity pruning technique to remove redundant branches from trees to maximize accuracy. Furthermore, the classifiers Radial SVM and KNN were used to measure the credibility of ensemble models by comparing the accuracy and kappa statistics. A repeated cross-validation was used as a training method. The results showed that the best performing algorithm was C5 with a kappa of 94.08%, accuracy of 97.04% and specificity of 96.47%, whereas SVM radical with a sensitivity of 98.08%. [13]. The purpose of Ullah's research was to establish a framework for analyzing network traffic flow data to classify IoT devices. The methodology approach consisted of a dataset with 80 network attributes. In addition, three K-fold "3, 5, and 10-fold cross-validation tests" was used to determine the possible validity and performance of the IoT system recognition methodology that we implemented for the IoT botnet dataset. A RFE technique was implemented to select significant attributes after extracting all correlated features from the IoT-AD-20 dataset. Furthermore, a random forest algorithm was used as an estimator by RFE for feature ranking. The RFE methodology ranks features based on their importance. The implemented model and dataset's validity were tested using accuracy, precision, recall, and F score. The Bmonitor device has precision of 99.0%, accuracy of 96.0%, recall score of 99.0%, and F1 score of 97.39%. [14]

V. METHODOLOGY

A. Data preparation

TABLE 2
LABELS FOR THE DATASET

Label	Description	Encoding
Benign	No malware detected	3
Malicious C&C	Command and control Attack where data is stolen from a device infected with malware	0
Malicious DDOS	The device is executing a distributed denial of service	1
Malicious PartOfAHorizontalPortScan	Collecting information from a device to prepare for a future attack	2

For data preparation, the Zeek files of both chosen captures are combined into a text file. The text file is imported to Microsoft Excel program and later imported into a csv file for processing. The initial set of features present in the dataset are shown in table 1 and the used set of features are marked in the column used of table 1. To clean the data, some features or columns were removed manually since all the data was missing. First, local_orig, local_resp, and tunnel_parents columns were eliminated manually from the set. Next, the columns for both label and detailed-label were combined

together into one label field which contains 4 different labels. Then, all labels are numerically one hot encoded as shown in table 2. All duplicate data was removed and the number of entries for each label was equally distributed and chosen. Additionally, the ts and uid columns were removed since it is not necessary to use the id assigned by the program or the time of entry of the packet. After cleaning and preparing the data, processing of the data is important. In order to use the IP address of the entries and obtain relevant classification results with, the IP addresses were converted into decimal format. To do this, the dotted decimal format is converted into 4 8-bit binary representations and finally combined and converted into decimal notation as shown in figure 1.

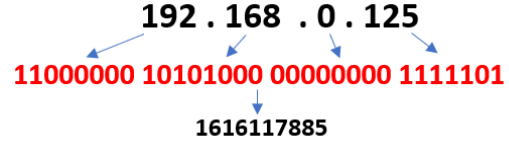


Fig 1. Decimal Representation of IP addresses in data

The features that are label encoded are: proto, service, conn_state and history. This done using sklearn label encoder functionality.

B. Splitting the Data

In order to use the data for ML processing, the dataset has to be split into testing and training sets. The training set will be used to train the model and the testing set will be used to analyze the performance of the model on unseen data. k-fold cross-validation will be used to split the data, where k=10 and k is the parameter that determines the number of groups to split the data into. First, the data will be randomized then split into k groups. Then, 9/10 of the groups will be used as the training set and the remaining group is used for the testing set. This will be done k times.

C. Classification models

The algorithms that will be used for this dataset to classify the different attacks will be Decision tree classifier, Naïve Bayes classifier, Random Forests, and Support Vector Machine (SVC) classifier. Decision trees are tree structures used for classification which contain decision nodes and leaf nodes. This algorithm will be used since it is inexpensive to implement, fast for classification of unknown records and can handle redundant attributes. After fitting the decision tree model on the data, the training and testing accuracy will be collected and used to determine whether the DT is overfitted, underfitted, or neither. The best size of the tree will also be determined by tweaking the DT hyperparameters. Naïve Bayes classifiers are based on probabilistic frameworks and are used in classification problems. The Bayes theorem will be used to compute the posterior probability, $P(Y | X_1, X_2, \dots, X_d)$. This algorithm will be applied since it is efficient and robust to isolated and irrelevant noise points. Additionally, the hyper parameters will be manually tweaked to make sure that the model is well-fitted. Support Vector Machine classifiers

(SVM) are supervised learning algorithms used mainly for classification problems and they work by categorizing the data, even if it is not linearly separable, by transforming data to a high-dimensional feature space. Linear SVMs will try to search for a decision boundary with the largest margin. The reason it will be used is that SVMs performs well on complicated domains. It will be used with linear and non-linear kernels and the training and testing accuracy will be reported. Random forests model will also be applied.

D. Performance Metrics

To evaluate the performance of our model on the data, different performance metrics will be used. Accuracy represents the number of correct classifications divided by the total number of classifications. The training accuracy will tell us about the accuracy of the model on the training data and the testing accuracy will tell us about the accuracy of the model on unseen data. Comparing the 2 accuracy measures will help in understanding if the model is overfitted or underfitted and therefore the steps to be taken to improve the accuracy of the model, such as increasing the size of the data or decreasing the number of features. Next, the confusion matrix, which is a table containing the representation of the instances of the predicted class and the actual class, will be constructed to help calculate important metrics for evaluation. The needed information from the confusion matrix is the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). The **precision** metric will allow us to examine the exactness and quality. This will be calculated by taking the ratio of true positives and total positives predicted. The **recall** metric is about the quantity and will be calculated by taking the ratio of true positives to all the positives. Since both precision and recall cannot be maximized at the same time, the F1 measure will be used to compute the harmonic mean of the two values. Having a high F1 will indicate both a high precision and recall. All plots will be generated to compare the models to each other and find the best model for this type of classification problem.

TABLE 1
ALL FEATURES INCLUDED IN DATASET

Feature	Description	Used
ts	Timestamp	0
uid	ID of connection	0
id.orig_h	The IP address of the sender or originator	1
id.orig_p	Originating endpoint's TCP/UDP port	1
id.resp_h	The IP of the receiver or Responding endpoint's	1
id.resp_p	Responding endpoint's TCP/UDP port	1
proto	Transport layer protocol of connection	1
service	Dynamically detected application protocol	1
duration	Time of last packet seen – time of first packet seen	1
orig_bytes	Originator payload bytes	1
resp_bytes	Responder payload bytes	1

conn_state	Connection state (13 connection states)	1
local_orig	If conn originated locally	0
missed_bytes	Number of missing bytes in content gaps	1
history	Connection state history	1
orig_pkts	Number of ORIG packets	1
orig_ip_bytes	Number of ORIG IP byte	1
resp_pkts	Number of RESP packets	1
resp_ip_bytes	Number of RESP IP byte	1
tunnel_parents	If tunnelled, connection UID of encapsulating parent	0
local_resp	the response originated local	0
Label	Benign or Malicious	1
Detailed Label	Type of Malicious entry	1

VI. RESULTS AND DISCUSSION

A. Decision Tree Classifier

After splitting the data using k-fold cross validation, the decision tree classifier was applied. The training and testing accuracy was obtained for every fold and the number of nodes expanded at every level was computed. When running the decision tree classifier, the maximum number of nodes that the model generated for this dataset was 34. The depth of the tree was 9. The number of nodes was plotted against the training and testing accuracy of each fold as shown in figure 2.

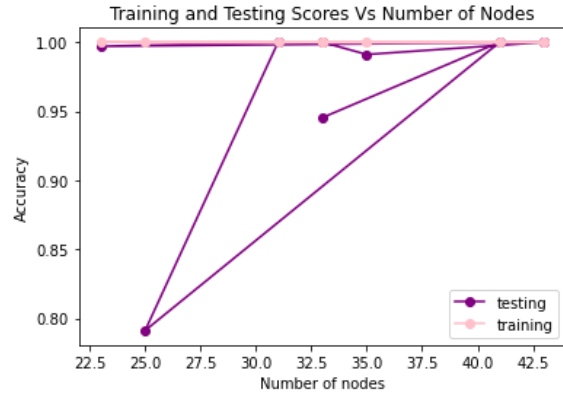


Fig 2. Training and testing accuracies vs node number

As can be seen in figure 2, the results show a general trend of a good model that is not overfitted or underfitted. The deviations in the range are due to the scale of the graph and the points plotted on the x-axis. The general decision tree graph can be seen in figure 3. To further test the Decision tree classifier, some hyperparameters were tuned. A max depth of 5 was applied to the model. It was observed that the best estimated number of nodes for our decision tree would be around 21 nodes for this parameter. The tree structure in figure 5 shows the results. The average accuracy was improved from “97.18 %” to “97.34%” and the overfitting is relatively reduced as shown in figure 4.

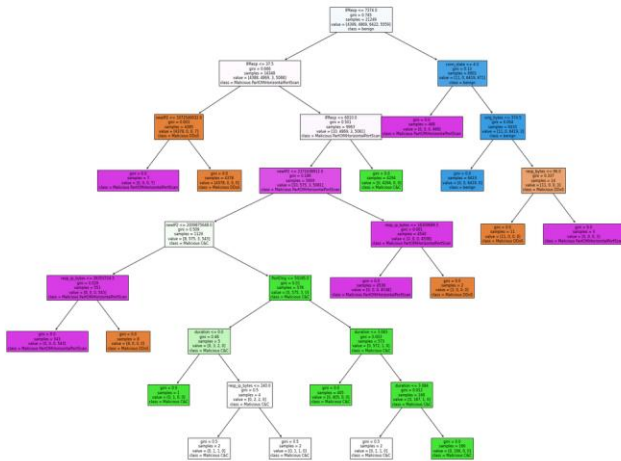


Fig 3. Decision tree graph

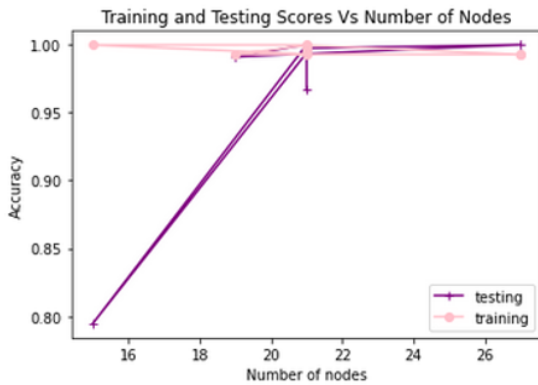


Fig 4. Training and testing accuracies vs node number

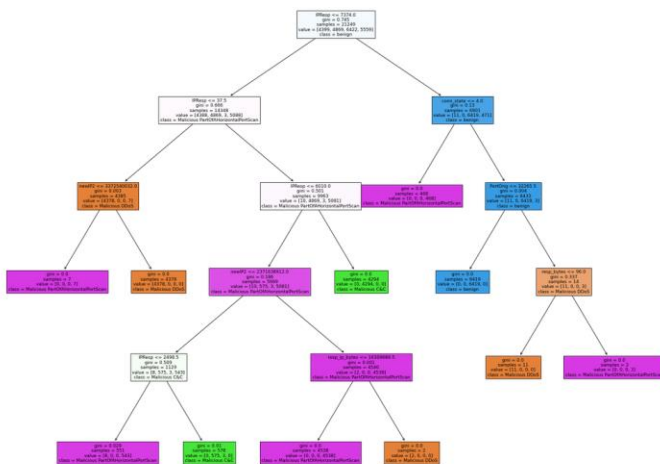


Fig 5. Decision tree graph for model after hypertuning

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